

ZYPER

Laser Ranged Delivery System

PROJECT ZYPER

ROLE: LASER RANGED DELIVERY SYSTEM

STATUS: UNDER DEVELOPMENT/ TESTING

ADVANCED AMATEUR ROCKETRY

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1. Introduction

Project “Zyper” (*Zip-er*) began as a thought experiment if a rocket could be made within a week before the start of an academic semester. Ideas were drawn and parts selected. It was thought reasonable since previous projects have used similar design construction/ methodology. Project Ketchup and SkyShot used a similar formulation of sugar rocket candy and casting method. However, there is a departure from using molded concrete nozzles which provided excellent impulse, and instead a 3-D printed nozzle. Other departures include the exclusion of the servo control system that was started in SkyShot. Lessons learned in dimensioning components to code development using the Arduino programming suite were hardened.

The ability to 3-D print parts in a precise fashion was a major limitation of SkyShot whose parts were cut from balsa wood, which severely limited making custom parts with the thickness and toughness needed.

2. System Components

2.1. Components

Commercial components purchased from websites such as Amazon, Ebay and spares from previous projects (Midnight Sceptre & Ketchup) were re-purposed for the use in Zyper. A list of components shown in Table 1 details the identification, function (and brief description) and cost. Where costs are not found a “X” is used in place. Only major components are listed in this table, smaller items such as the custom 3D prints and ignitors are listed in the “*Figures / Data / Components*” section of this project paper.

Table 1: Component listing and their descriptions.

Identification	Function (Desc.)
VL53L1X	Laser range finder, 400mm capability, +/- 0.02mm accuracy (reported).
Amazon Basics 9V (2)	Power to Arduino and ignition of EP.
Arduino Uno	Micro controller, takes data to perform objective.
MPU 6050	Accelerometer, acts as 1 st safety switch upon G threshold.
Ignitor (Estes Brand)	Commercial model rocket ignitors.

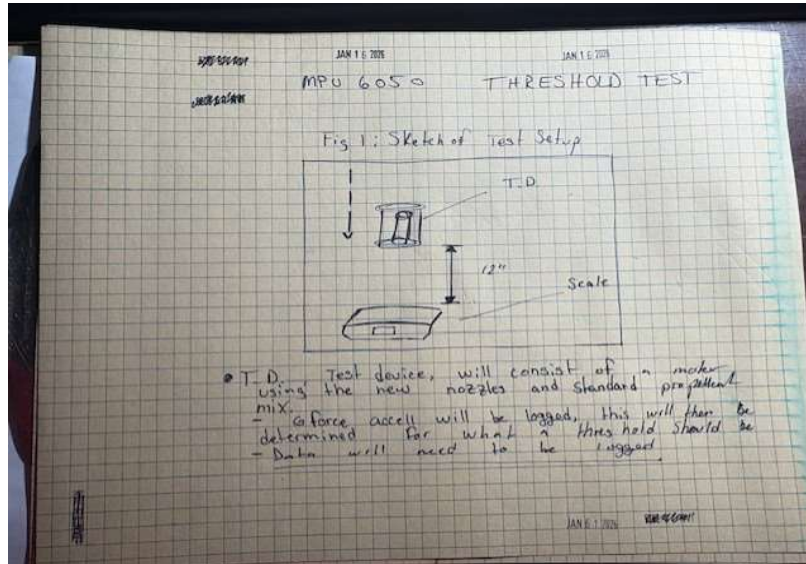
2.2. Component Characteristics

Items selected for this project were chosen off of the following criteria, availability, cost and ease of final assembly. Briefly, a discussion on some items such as the system safeties and the battery wiring are discussed.

Threshold value for the acceleration safety was determined by way of using a incomplete mock-up of the full scale test version of Zyper. This scaled down test was used to measure the G-Force imparted onto the MPU6050 in a horizontal launch scenario- which mimics the conditions that would be encountered. Components used in this test were the MPU6050, Arduino Uno, **NA-MicroSD data logger** and a power source (9v). The test configuration was arranged so that the G-Force could be measured in a live fire demo and the force imparted onto the scale so that the curve could be integrated and the impulse found for a partially loaded Zyper system. An

illustration of this arrangement is shown in Figure 1¹. The battery module consists of 2 9v batteries whose wiring configuration is a series combination (Figure 12).

Figure 1: Illustration drawing of test configuration.



2.3. New Design Introduction

Certain elements present such as the 3-D printed ablative rocket nozzle and the Explosive Package (EP) are new and their testing in a working environment is limited either due to time or cost complexity. Small tests of the 3-D printed nozzle with and without a 1mm graphite-glue coating has been done to test to see whether a performance difference could be quantified. Further discussion of these items is presented in the following sections.

2.3.1. Laser Range Detonation Mechanism (LRDM)

The large availability of commercial modules that can be purchased and provide accuracy of more than a centimeter made it attractive to incorporate it into a project with solid rocket propulsion. The chosen Time of Flight sensor (or LRDM) was the VL53L1X. A class one laser of 940nm made selection ideal as ~94% of transmission through acrylic occurs²[6]; suitable for the purpose of this to protect the sensor in flight and transport. Iteration 1 of the LRDM housing and the “*mounting well*” where the sensor would be mounted centrally in the 3D print

¹ A test was devised but not carried out* G-Force trigger set to 10G, which upon MPU6050 testing is sensitive enough to move forward.

² Testing as of 1/19/25 indicates that range is limited to practical 200mm** with acrylic dome in iteration 1 configuration.

Figure 2: LRDM Housing (*Pre-fitment*). Iteration 1.

The timing budget of this sensor is also ideal as this is the maximum time allowed for the sensor to capture readings; of which, the LRDM has a 20ms timing budget. The device also has a user adjustable single photon avalanche diode (SPAD) array with integrated lens which is capable of changing the field of view (FOV) of the device. The default SPAD 16x16 was used in tests and no calibration was performed to correct for the acrylic casing located in front of the LRDM. Multiple design iterations were made on the sensor mounting as documentation regarding the mounting and “crosstalk” effect from having an airgap with a protective material overhead the sensor could inhibit reliable readings. Initial measurements with the design enclosure and mounting (Figure 2) well produced measurements results not past 210mm. Measurements taken are also averaged at 10 (*1 measurement point = 10 true measurement points averaged*) for the code.

Figure 3: Device FOV and Adjustable Region of Interest (ROI) on sensor capability. (27deg actual FOV)

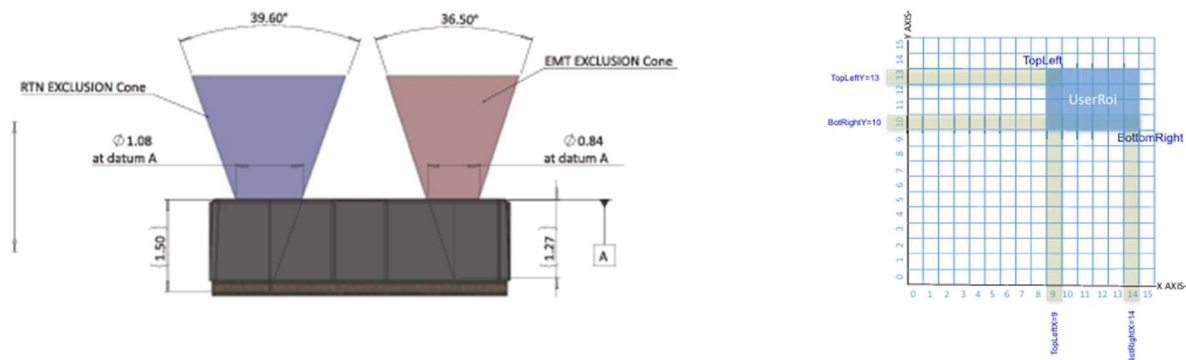


Figure 4: Approximate size of VL53L1X.

2.3.2.3-D Printed Nozzles

Nozzles that were previously used in SkyShot could not be replicated. The mold used to create them was lost and a substitute was needed³. Single tests were conducted with graphite and Elmers glue mixture at a ratio of ____/____ and compared to the standard print.

Table 2: Results from single shot experiments.

With Ablative coating (AC)	Standard Print (SP)	Impulse (Calculated)
		(AC)
		(SP)

Final fitment of the nozzles were placed into a PVC Schedule 40 (~5in) long propellant body with an end cap and nozzle shroud. Figure 5 showcases the exploded view of these components.

³ Previous nozzles were purchased from a kit, satisfactory results from 3-D experiments proved that printed ones could achieve similar outcomes.

Figure 5: Exploded view of the solid rocket booster.

2.3.3. Explosive Package (EP)

To differentiate this project from previous ones the parts were printed and a laser range finder to trigger a primary and secondary charge for an explosive payload was done. Figure 5 briefly describes this subsystem. More information of the system can be found in Section 5 “Figures/ Data/ Components”.

The “Primary” and “Secondary” descriptions given in Figures 6 & 7 are misleading as the Primary charge is the activator of the more damaging materials located in the secondary charge. The primary is filled with crushed sparkler dust, connected to a Estes model rocket ignitor. Upon receiving the previous safety checks and a proper laser range the secondary is ignited almost instantaneously. The secondary consists of liquid gasoline. The compartments for the secondary and primary are sealed preventing liquid and sparkler dust mixing and escaping prior to being triggered. A close up of this secondary charge where the ignitor and sparkler dust would be housed can be seen in Figure 7.

Figure 6: Illustration of the EP and the Primary/ Secondary explosion stages.

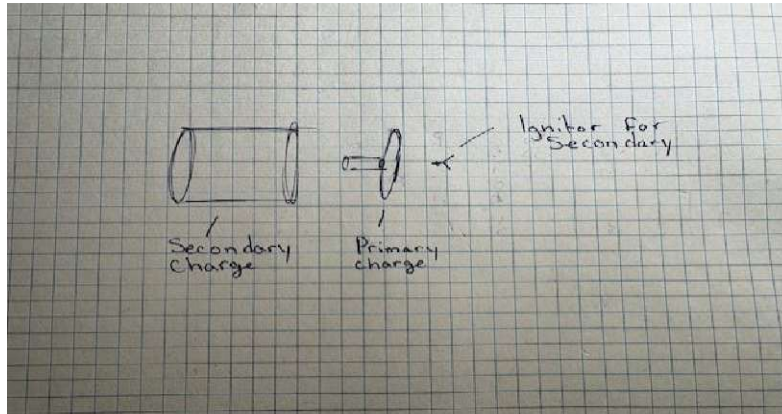


Figure 6a: Image of Primary charge (Red), Secondary Charge (Blue), Ignitor (Orange) and Battery.

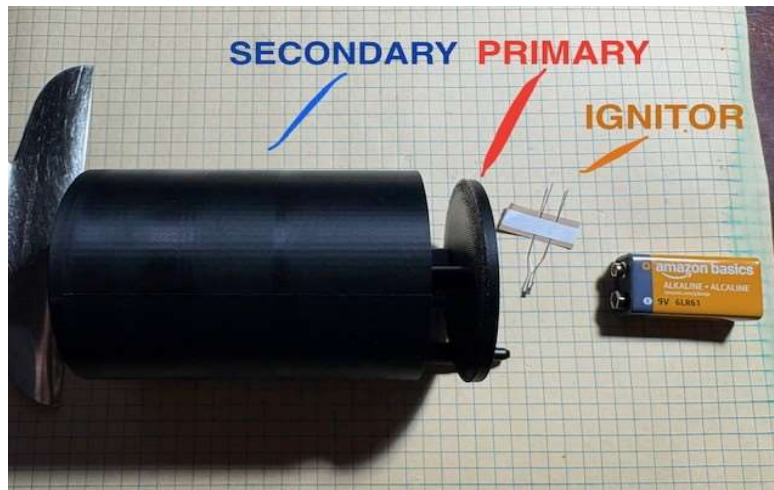
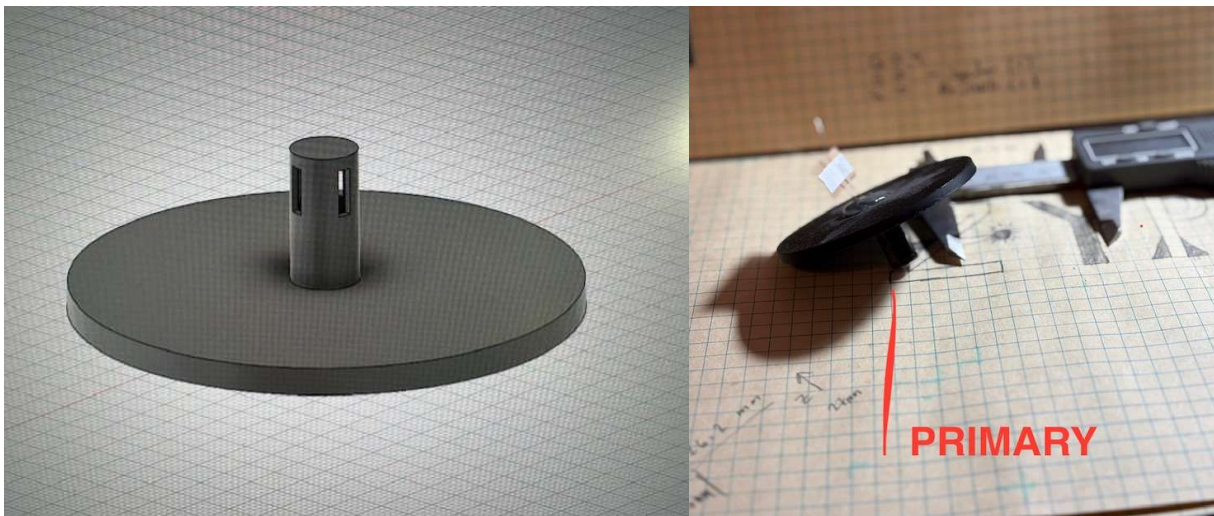


Figure 7: Closeup image of Primary ignitor.



2.3.4. Rocket Fin Housing

The final addition that would differentiate this from previous projects is the single print fin assembly. The sleeve and fins are attached to slide into the cardboard tube with a PVC motor housing for propulsion. Presented images are CAD models of the various fins used.

Figure 8: Assorted Fin configurations used.

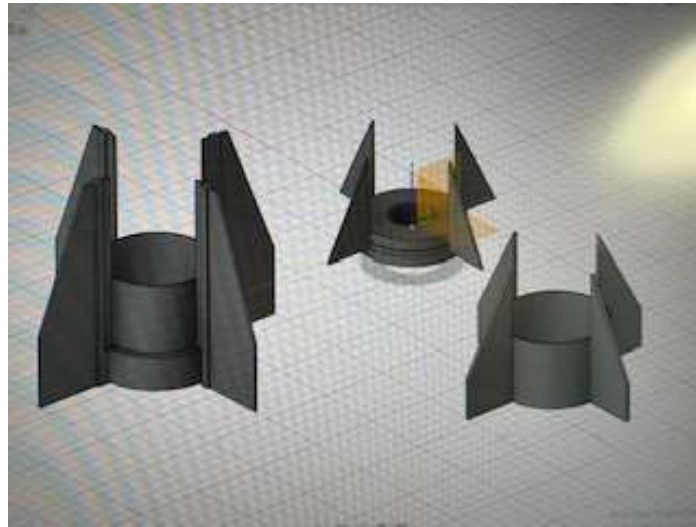


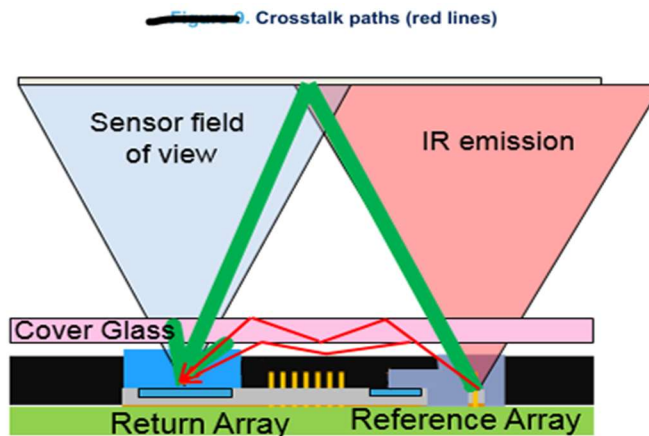
Figure 8a: Physical fin attachments.

NA

3. Limitations and Challenges

3.1. Cross talk phenomenon

Initial understanding of the LRDM sensor was that It would be adequate in the first iteration enclosure, after testing to verify max range and repeatability of range measurements it was found that the air gap between the sensor and acrylic border inhibited range functionality [6]. Iteration 1a incorporated a mount for a small camera to record in flight video and slightly raised the mounting well into a pedestal for the LRDM sensor, range results improve marginally with this addition. Iteration 2 consisted of the raised pedestal to within 2mm of the dome acrylic housing as recommended per manufacturer spec. Figure 9 showcases the LRDM mount iterations and Figure 10 the documentation regarding the “crosstalk” phenomenon. LRDM mounting pedestal height was increased to reduce cross talk, and the com wire well was widened for ease of access.

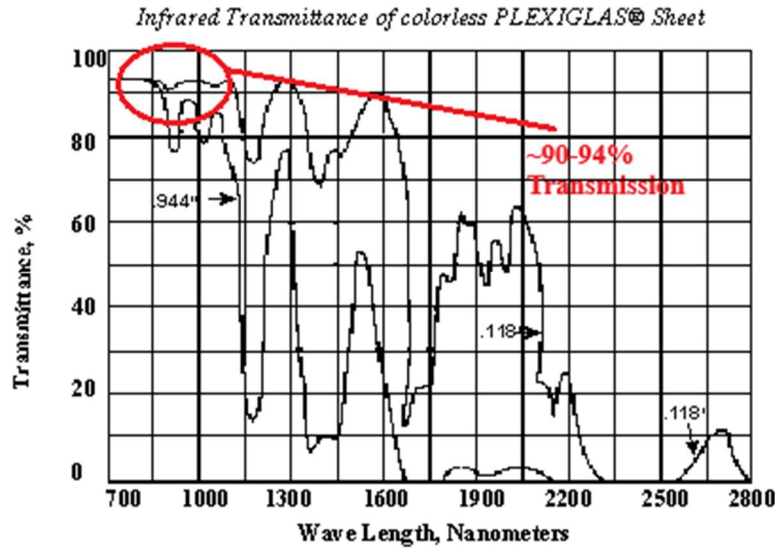
Figure 9: Images of Iteration 1, 1a and 2.**Figure 10: Cross Talk phenomenon per manufacturer spec.**

This “cross talk” occurs when the interference from reflection of a cover material and not the intended target which distorts data⁴. The air gap from the LRDM and the cover material as well as the thickness of the cover material increases these reflections⁵. As the protective dome purchases was through a generic supplier on Amazon true composition cannot be known, however brand name acrylic- Plexiglass has data regarding optical transmission for clear plexiglass. Clear plexiglass is regarded as acrylic for the information presented in Figure 11.

⁴ It should be noted that other contributors of cross talk are; smudge, angle of cover material mounting and other Infrared light sources.

⁵ NX-LRPD-01-b was given a 3-D printed dome enclosure with a portlet at the top to increase range readings. Increasing the pedestool height within 1mm of acrylic dome only slight increased sensitivity and range readings.

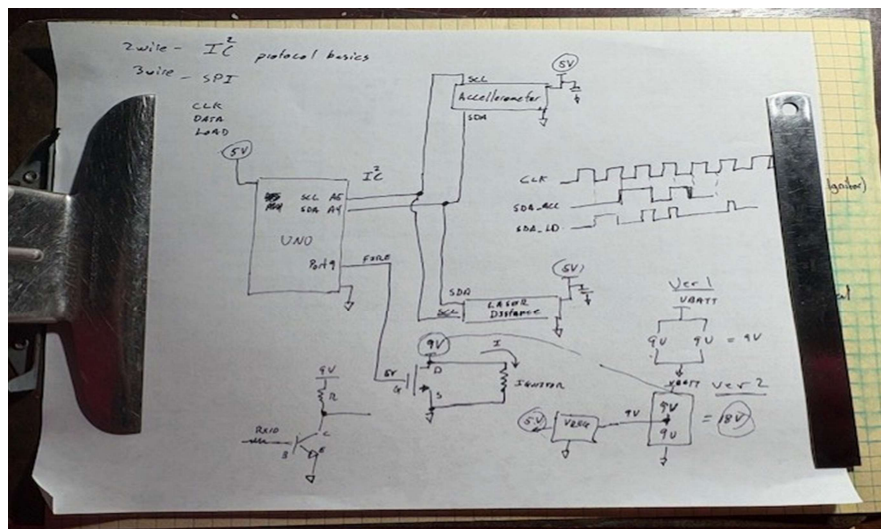
Figure 11: Optical Transmission of Clear Plexiglass (Acrylic)



3.2. Circuit Design

When this project was conceived and written the idea was simple to have multiple safeties and a laser ranging device to trigger a controlled explosion it seemed simple. Challenges encountered early on with the switch to deliver sufficient current to the ignitor to ignite the EP were discovered. Use of a N-Type MOSFET was used to get around this and deliver the full amperage to the ignitor from the battery bank was done. While not tested, it is the suspicion that the use of heavier gauge wires would reduce the resistance and deliver a faster and hotter burning igniter. Figure 12 with the circuit in question designed for this project. Further improvements made the circuit design would include a physical switch to turn on the Arduino and an indicator light for visual cue that the system is primed.

Figure 12: Project Zyper circuit.



4. EXECUTIVE SUMMARY: PROJECT ZYPER

The goal of developing a small-scale rocket whose on board electronics could trigger an explosive package was met, design of a circuit and system able to deliver is considered achieved; however, not without significant limitations and revisions made. Use of a commercially available TOF sensor, knowledge in sugar rocket casting, 3-D Printing and circuit design aided in the construction and assembly of this system. Further refinements to the system would include, further measurement of the timing budget between MPU6050, TOF sensor, and Arduino UNO, inclusion of the solid caste nozzle, and investigating the possibility of additional/ alternate TOF sensors.

5. References

- [1] “MPU-6000 and MPU-6050 Product Specification Revision 3.4”
- [2] “UM2356 User manual VL53L1X API user manual”
- [3] “A new generation, long distance ranging Time-of-Flight sensor based on ST's FlightSense technology”
- [4] “FAIRCHILD SEMICONDUCTOR, RFP22N10, RF1S22N10SM, DataSheet, Jan 2001.”
- [5] “Arduino, Arduino® UNO R3 User Manual”
- [6] “Cover window guidelines for the VL53L1X long-distance ranging Time-of-Flight sensor”

6. Figures/ Data / Components

Figure ____: _____.

Figure ____: _____.